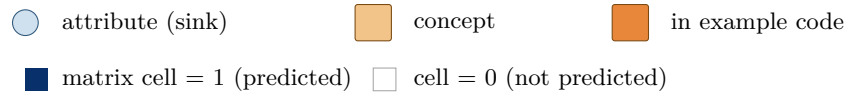


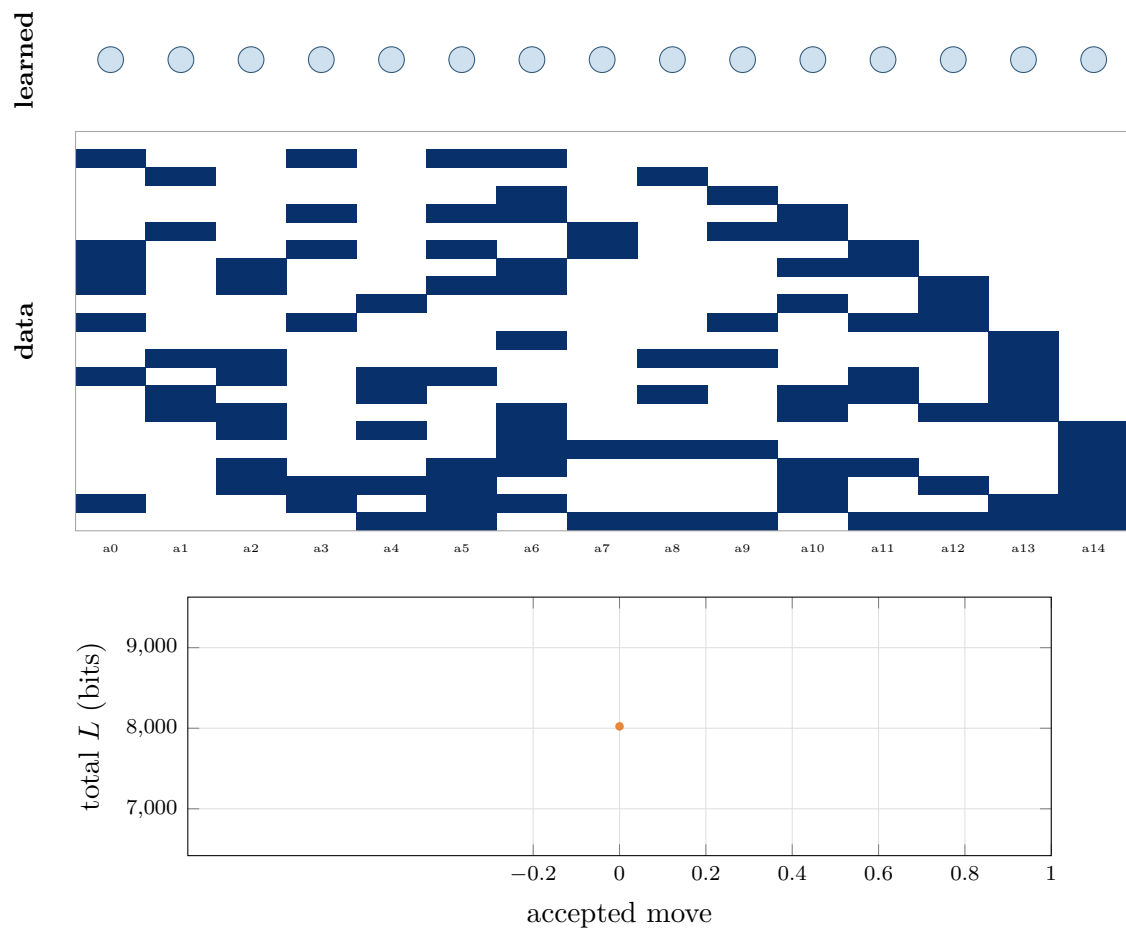
mdl-fca demo figures

The learner builds a concept DAG that compresses a binary object $\times$ attribute matrix, positing a concept only when it shortens the total description length. Figures are generated as native TikZ from the learner's output.



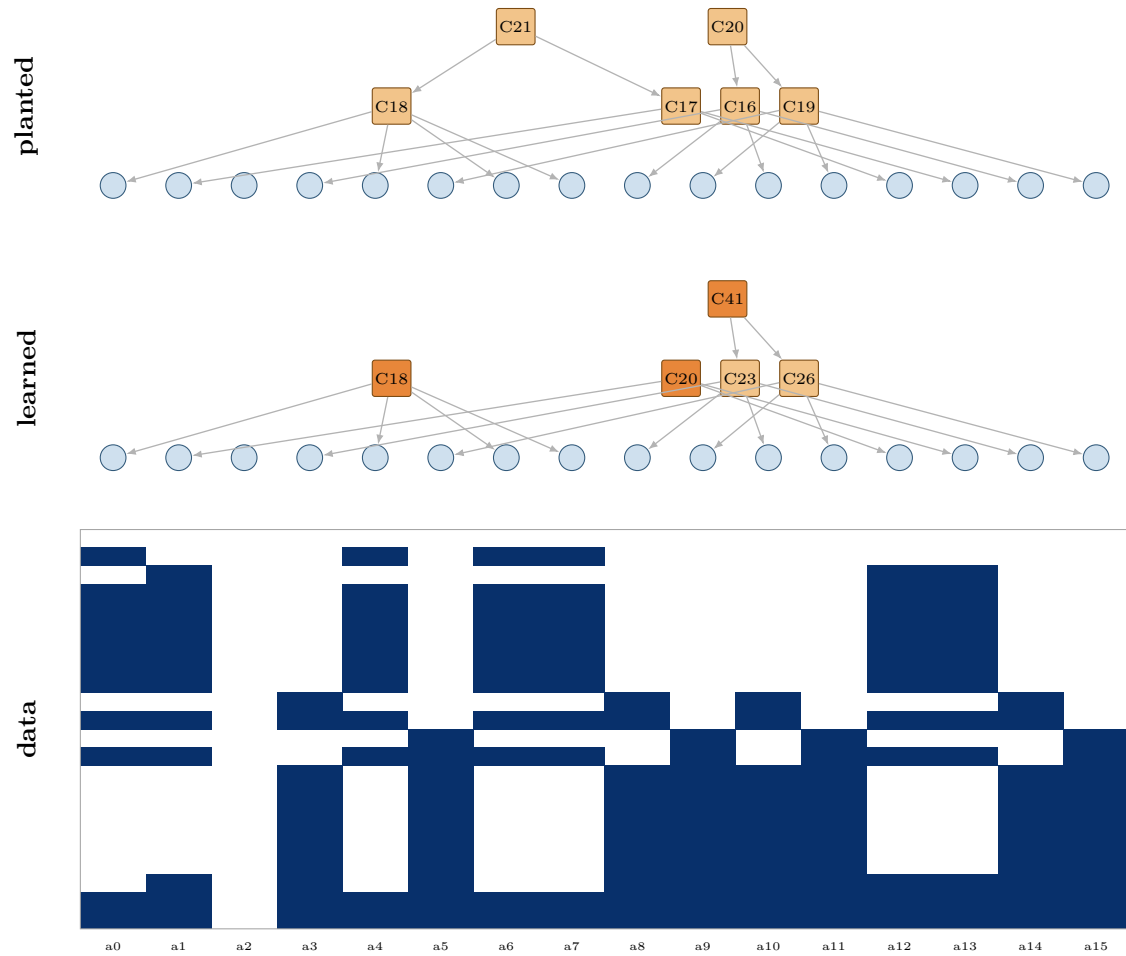
Control: uncorrelated noise

Independent Bernoulli(0.3) with no hidden structure. Nothing is a suspicious coincidence, so no concept pays for itself: the learner returns **0 concepts**. The control for over-generation.

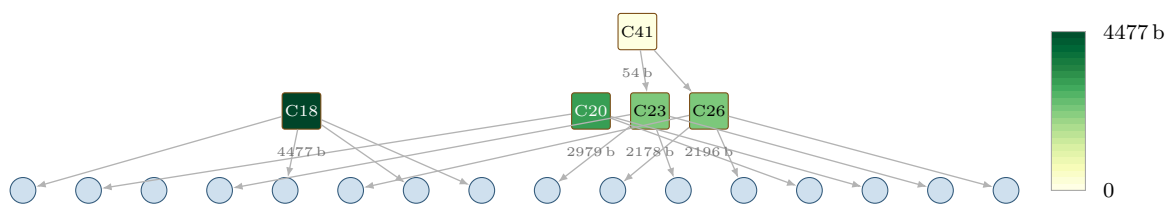


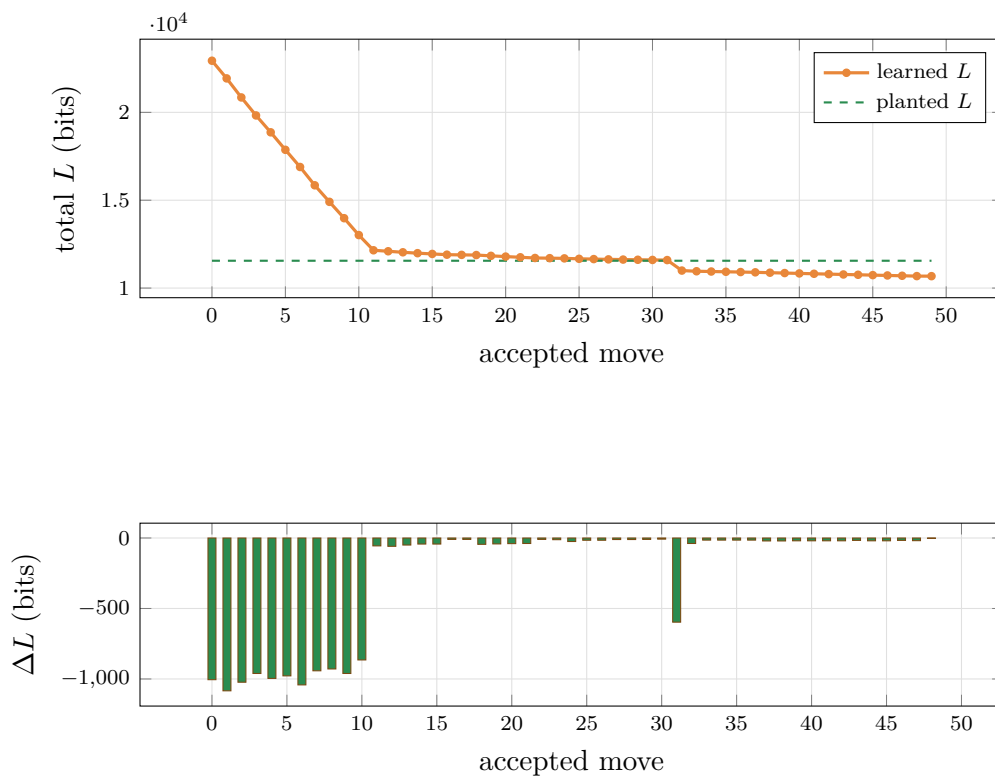
Planted two-level hierarchy

Four base concepts under two parents. Recovered, and compressed below the codelength of the generating model.



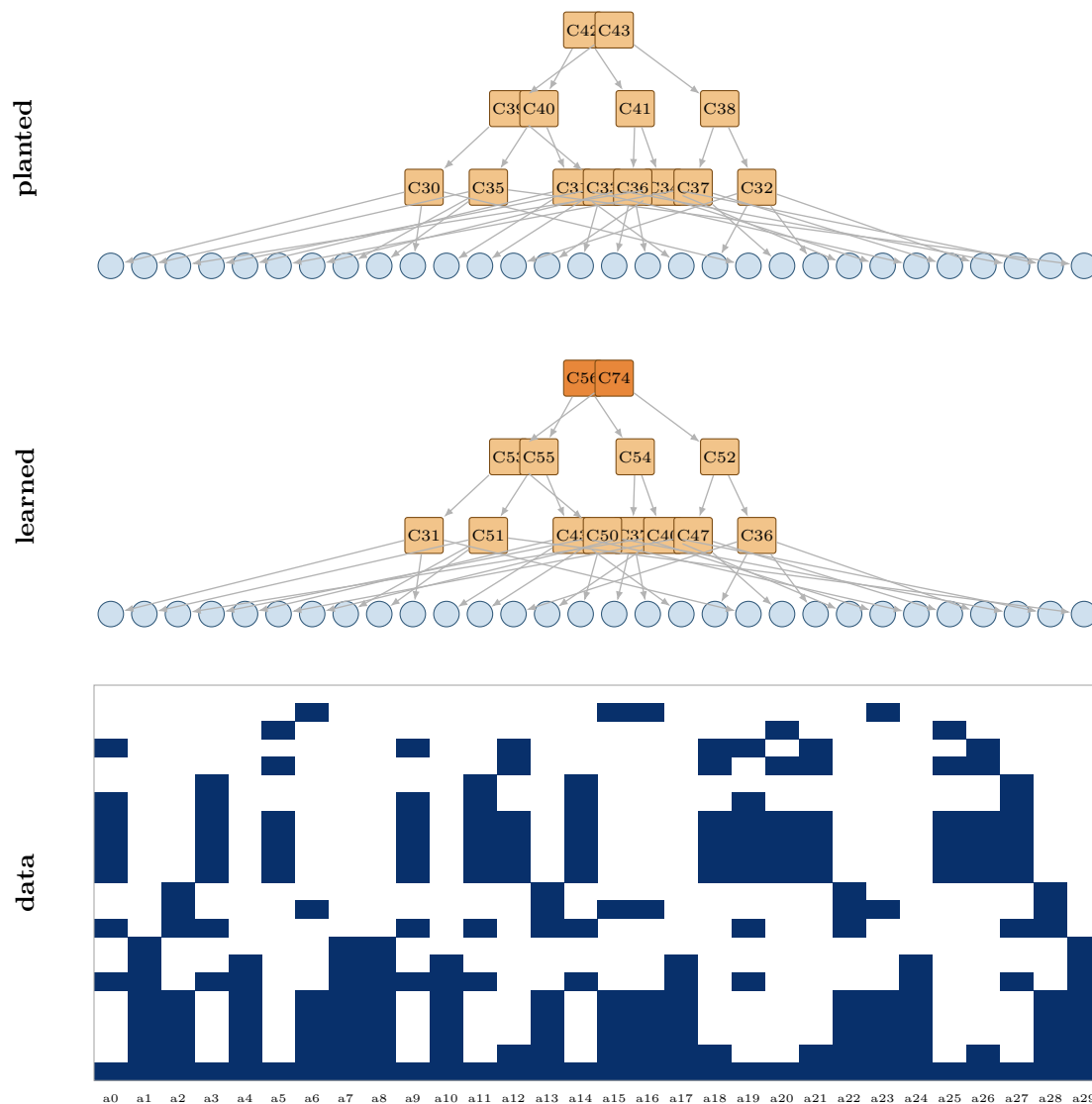
Rent (bits lost if a node were removed; node colour):



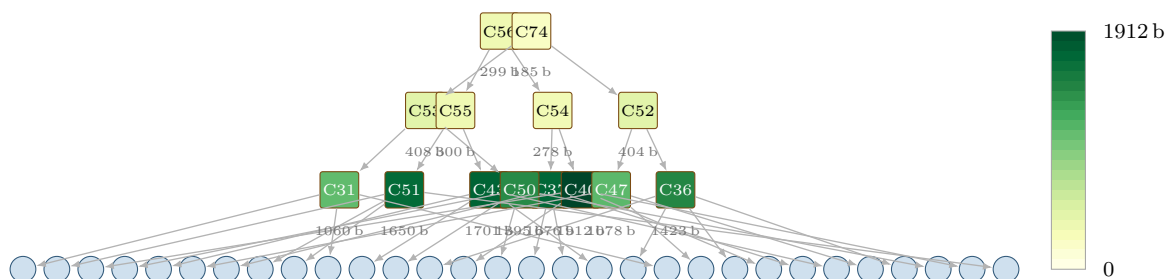


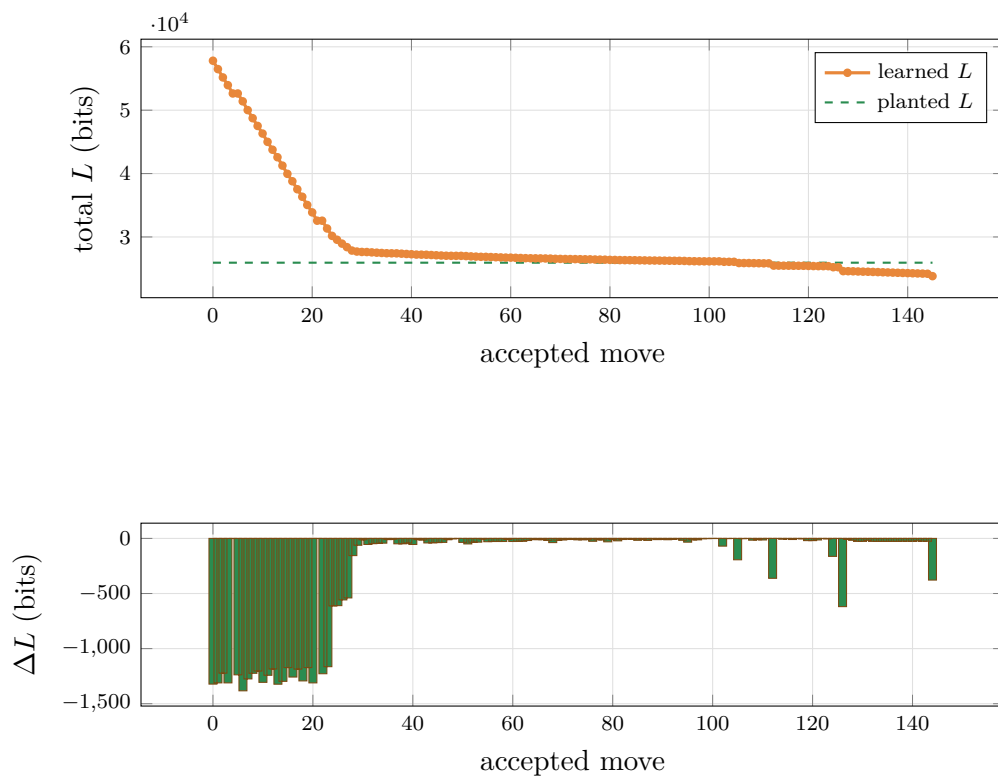
Planted three-level hierarchy

Base $\rightarrow$ mid $\rightarrow$ top. All concepts recovered in three tiers that align with the planted ones column-for-column; base concepts carry most of the rent.



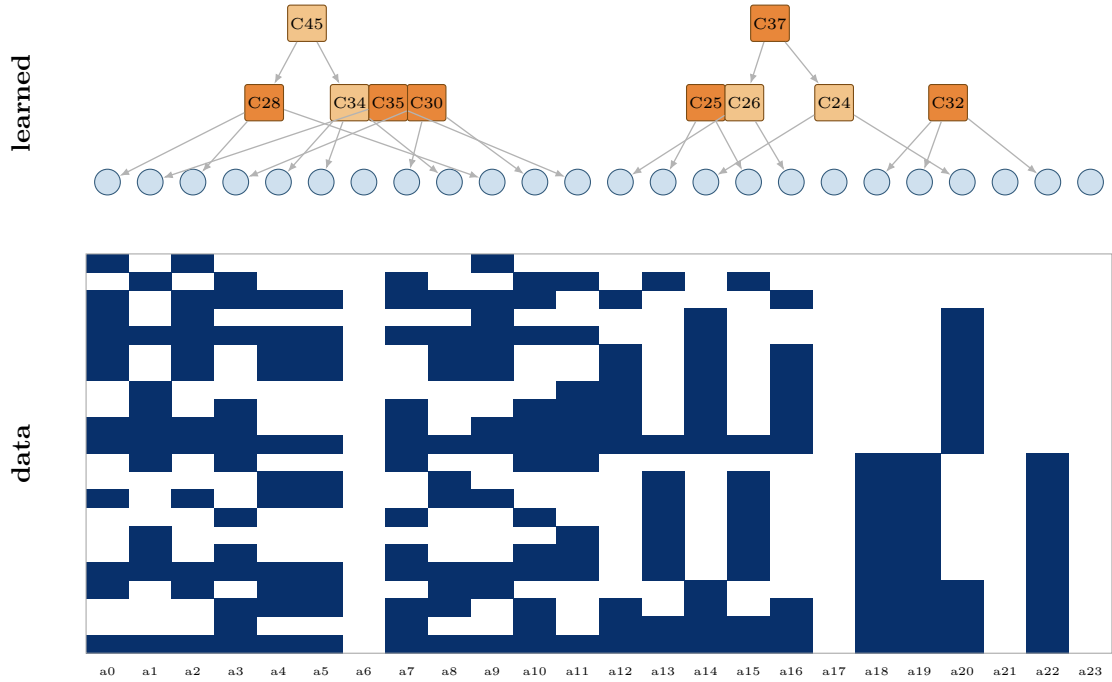
Rent (bits lost if a node were removed; node colour):





Two independent factors (a product)

Two independent hierarchies on disjoint attribute blocks, concatenated. The learner returns two disconnected components with no cross-block concept; the product lives in the object codes.



Rent (bits lost if a node were removed; node colour):

